An open benchmark suite and reference baselines for high-entropy alloy phase prediction

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Abstract

Empirical phase-prediction rules for high-entropy alloys are widely used as fast pre-screens for candidate compositions. The four most common rules are the entropy criterion of Yeh and colleagues [1], the atomic-size mismatch criterion of Zhang and colleagues [2], the valence-electron-concentration criterion of Guo and Liu [3], and the Ω-parameter criterion of Yang and Zhang [4]. The accuracies reported in the original publications, however, were obtained on heterogeneous and mutually incompatible datasets, and no published artifact treats the four rules as diagnostic classifiers on a common open benchmark. This work introduces HEA-BENCH, an open Python library and reproducible benchmark suite that consolidates three primary open phase-label datasets into a single 7784-composition resource with explicit per-row source provenance and cross-source label-conflict tracking, implements the four rules as binary or multi-class diagnostic classifiers with full statistics (accuracy, sensitivity, specificity, Wilson 95% confidence intervals, receiver-operating-characteristic curves, and Youden's J), and ships in two equivalent user surfaces. The first is a standard installable Python package. The second is an in-browser frontend that runs the same release wheel through Pyodide on WebAssembly, requiring no local installation. On the consolidated benchmark, both binary rules collapse to a "predict-single-phase-almost-always" regime with Youden's J values of 0.075 and 0.032, and a sweep of the Y. Zhang et al. [2] δ threshold places the dataset-optimal balance point at $\delta < 2.5\,\%$ rather than the canonical $\delta < 6.5\,\%$. The package is released as version 0.0.1 under the MIT License with more than 150 tests, including pinned regression tests on every reported benchmark statistic.

Keywords: high-entropy alloys, multi-principal element alloys, phase prediction, benchmark suite, empirical descriptors, Miedema model, open data, diagnostic classifiers

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Required Metadata

1. Motivation and significance

High-entropy alloys, introduced independently by Cantor and colleagues [5] and by Yeh and colleagues [1] in 2004, are multi-component alloys with five or more principal elements near equiatomic composition.

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Table 1: Code metadata.

Nr.	Code metadata description
C1 Current code version	0.0.1
C2 Permanent link to code/repository	To be assigned via Zenodo on the first tagged release
C3 Permanent link to reproducible capsule	Not applicable. The package is dependency-free in its core and reproducible via a standard package-manager install.
C4 Legal code license	MIT
C5 Code versioning system used	git
C6 Software code languages, tools, and services used	Python ($\geq$3.10), Pyodide for browser deployment
C7 Compilation requirements, operating environments, and dependencies	Pure Python, dependency-free core. Optional extras for tabular data handling, plotting, and testing.
C8 If available, link to developer documentation or manual	Repository readme file
C9 Support email for questions	`davjfies@gmail.com`

The configurational mixing entropy in such alloys is large enough to stabilize disordered solid solutions over the intermetallic compounds predicted by classical Hume–Rothery rules. The compositional space is combinatorially larger than that of conventional alloys, which makes pre-screening a practical necessity before any expensive density-functional-theory, CALPHAD, or experimental campaign. The practical importance of robust pre-screening is illustrated by recent work on the thermodynamic stability of femtosecond laser-ablated $CuCoMn_{1.75}NiFe_{0.25}$ nanoparticles [6], in which the choice of phase-prediction descriptor materially affected the interpretation of stability during the surface restructuring process.

Four empirical phase-prediction rules dominate the high-entropy alloy literature. The first is the configurational mixing entropy criterion of Yeh and colleagues [1], which classifies an alloy as high-entropy when $\Delta S_{\mathrm{mix}} > 1.5R$. The second is the atomic-size mismatch criterion of Zhang and colleagues [2], which predicts a single-phase solid solution when $\delta < 6.5\,\%$. The third is the valence-electron-concentration criterion of Guo and Liu [3], which predicts a face-centred-cubic structure at VEC ≥ 8.0 and a body-centred-cubic structure at VEC < 6.87. The fourth is the Ω-parameter criterion of Yang and Zhang [4], with $\Omega = T_{\mathrm{m}}\,\Delta S_{\mathrm{mix}}\,/\,|\Delta H_{\mathrm{mix}}|$ and a threshold of $\Omega > 1.1$ for single-phase stability. Each rule was supported in its original publication by classification accuracies near 80 % on the dataset available at the time.

Three gaps in the open literature motivate the present work. First, every published evaluation of these rules uses a different dataset, with different preprocessing, deduplication, and phase labelling. Head-to-head comparison across rules and across methods is therefore impossible. To the author's knowledge, no open benchmark suite for high-entropy alloy phase classification exists that is analogous in scope to the MatBench resource for materials property prediction [7]. Second, the rules are typically reported as accuracies of the form X/N, without sensitivity, specificity, Wilson intervals, receiver-operating- characteristic analysis, or Youden's J. This omission obscures the regimes in which each rule actually generalizes. Third, the only widely used interactive tool for Miedema-style enthalpy calculations is the registration-gated, Windows-only MiedCalc program of Zhang and colleagues [8]. No zero-install open analogue is currently available.

The contribution of this work addresses all three. It consolidates three primary open phase-label datasets into a single resource with full provenance. It implements the four canonical rules as diagnostic classifiers and reports their statistics on the consolidated benchmark. It distributes the result in two equivalent surfaces, namely a standard Python library and an in-browser front end that executes the same release wheel through the Pyodide runtime.

2. Software description

2.1. Software architecture

The package is organized as a small set of orthogonal sub-components and a thin command-line interface. A composition component performs formula parsing and normalization, with a single regular-expression-based parser that accepts the three formula conventions found in the source datasets, namely space-separated proportional amounts, compact element-coefficient strings, and bare formula tokens. A descriptor component implements the six empirical descriptors required by the rule set, namely ΔS_{mix}, δ, the valence electron concentration, T_{m}, ΔH_{mix} [9, 10], and Ω [4]. Per-element parameters covering 24 elements are held in an internal data table. Binary mixing-enthalpy pairs are loaded from a vendored copy of the matminer Miedema pair table [11], licensed under BSD-3-Clause with the upstream license file preserved alongside the data. The vendored table derives from the de Boer parameter set [12] and is consistent with the open tabulation of Takeuchi and Inoue [13].

A rules component holds the four canonical empirical rules wrapped as classifiers, each with a tunable threshold defaulting to the value reported in the original publication. A classifier-statistics component provides pure-stdlib implementations of Wilson score intervals [14], confusion matrices, binary classifier statistics, and receiver-operating-characteristic sweeps with Youden's J [15]. A benchmark component holds the per-source loaders, the cross-source consolidator, and a coverage analysis that quantifies which fraction of the consolidated dataset is scorable by the current descriptor data tables. An evaluation orchestrator runs each rule against the consolidated benchmark and writes the resulting statistics as a structured release artifact.

Two end-user surfaces consume the library. The Python surface is a standard package-manager-installable distribution. The browser surface is a single HTML page that loads the Pyodide runtime [16] and, in turn, installs the same release wheel that the package manager would deliver. Descriptor and rule calls then execute client-side in the user's browser tab, with no server-side component. The library has one canonical implementation, and both surfaces invoke it identically.

2.2. Software functionalities

The package supports six primary workflows. A user can parse an arbitrary composition formula, normalize it to mole fractions, and compute the six standard descriptors. A user can apply each of the four canonical empirical rules and obtain a predicted phase classification. A user can score a user-supplied rule or composition set against the v0.1.0 benchmark with full diagnostic statistics. A user can sweep a rule threshold and obtain the receiver-operating-characteristic curve, the optimal-accuracy threshold, and the optimal Youden's-J threshold. A user can analyse element coverage of an alloy set against the descriptor data tables and inspect a ranking of next-best-add elements. A user can re-consolidate the per-source data through the bundled loaders and the deduplication and conflict-detection logic.

2.3. The consolidated benchmark

Version 0.1.0 of the package ships a single canonical benchmark containing 7784 unique compositions (Figure 1). It is built by running the per-source loaders on three open datasets. The dataset of Borg and colleagues [17] contributes 740 unique (formula, processing) alloys, licensed under CC-BY-4.0 and mirrored verbatim with a recorded SHA-256. The dataset of Pei and colleagues [18] contributes 1209 unique formulas, also CC-BY-4.0 and mirrored. The dataset of Peivaste and colleagues [19] contributes 7747 unique formulas after deduplication of an upstream-generated combinatorial sweep. The upstream data repository for the Peivaste dataset declares no license and is therefore not redistributed in the package, but is fetched on user request at build time.

Compositions are merged on a four-decimal-precision composition key. The processing column from the Borg dataset is preserved as a side-channel rather than included in the join key, in order to keep the canonical taxonomy composition-only. When two sources assign different canonical labels to the same composition, the row is flagged and the per-source labels are preserved verbatim. The v0.1.0 release contains 100 such cross-source conflicts, which represent 1.3 % of the benchmark. These conflicts are themselves a useful artifact, because they enumerate compositions on which the open high-entropy alloy literature does not yet agree.

The canonical phase taxonomy is deliberately coarse for v0.1.0. The four classes are body-centred cubic, face-centred cubic, hexagonal close packed, and multi-phase. This coarseness allows all three sources to map into the canonical scheme without lossy guessing. Finer per-source labels are preserved as side-channel columns. These include the microstructure column from the Borg dataset, which carries more than thirty sublabels, and the twelve-category column from the Peivaste dataset, which distinguishes intermetallic and amorphous phases.

3. Illustrative examples

3.1. The Cantor alloy as a sanity check

The canonical equimolar CoCrFeMnNi alloy of Cantor and colleagues [5] serves as the field's reference high-entropy alloy and as a regression sanity check for the library. The six descriptor values returned by the

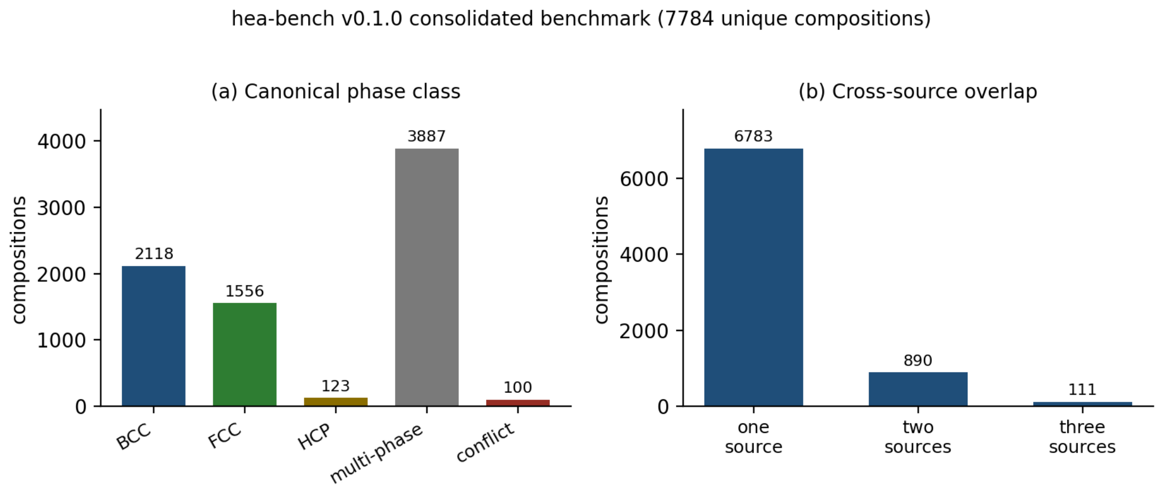

Figure 1: Composition of the v0.1.0 consolidated benchmark.

package for the equimolar composition are $\Delta S_{\mathrm{mix}} = R \ln 5 = 13.381\,\mathrm{J/(mol \cdot K)}$, $\delta = 3.164\,\%$, $\mathrm{VEC} = 8.000$, $T_{\mathrm{m}} = 1801.2\,\mathrm{K}$ (rule-of-mixtures average), $\Delta H_{\mathrm{mix}} = -4.16\,\mathrm{kJ/mol}$, and $\Omega = 5.794$. All four empirical rules classify the Cantor alloy as high-entropy class, single-phase, and face-centred cubic, in agreement with the original experimental characterization [5]. Each of the six values above is pinned in the regression test suite.

3.2. Reference rule baselines on the consolidated benchmark

Running the four rules through the evaluation orchestrator produces the reference baselines shown in Table 2. The reported values are pinned by the test suite. Any future change in the dataset, the descriptor implementations, the vendored pair-enthalpy table, or the rule thresholds will perturb these values and will also break the build.

Table 2: Binary-classifier baselines on the v0.1.0 benchmark.

Rule	n	Accuracy	Sens.	Spec.	J
$\delta < 6.5\,\%$ [2]	6651	56.7 %	99.0 %	8.5 %	0.075
$\Omega > 1.1$ [4]	6651	54.4 %	95.8 %	7.4 %	0.032

The Wilson 95% confidence interval on the Zhang δ accuracy is $[55.5\,\%, 57.9\,\%]$, and on the Yang Ω accuracy is $[53.2\,\%, 55.6\,\%]$. The two binary rules return Youden's J values close to zero, which indicates that on the consolidated benchmark they collapse to a regime of predicting single-phase almost always.

The VEC rule of Guo and Liu [3] predicts crystal structure rather than single-phase formation, and is therefore evaluated by restricting observations to those labelled face-centred-cubic or body-centred-cubic. On the 3463 single-phase observations in the benchmark, the rule attains an accuracy of 66.9 %. The per-class breakdown shows a strong asymmetry, with 92.3 % of observed face-centred-cubic alloys predicted correctly but only 48.3 % of observed body-centred-cubic alloys.

The Yeh and colleagues [1] ΔS_{mix} criterion is descriptive rather than predictive. On the consolidated benchmark, 47.0 % of compositions pass the $\Delta S_{\mathrm{mix}} > 1.5R$ high-entropy threshold, 36.9 % fall in the medium-entropy range $R \leq \Delta S_{\mathrm{mix}} \leq 1.5R$, and 16.1 % fall below the medium-entropy threshold.

3.3. Threshold recalibration of the Zhang rule

Treating the rules as diagnostic classifiers admits a question that the canonical literature does not address. What threshold would maximize Youden's J on this benchmark? Sweeping the Zhang [2] δ threshold from 1 % to 12 % in steps of 0.1 % yields the result summarized in Table 3.

Table 3: Receiver-operating-characteristic sweep of the Zhang δ rule on the v0.1.0 benchmark.

Threshold	Sens.	Spec.	Youden's J
6.5 % (canonical)	99.0 %	8.5 %	0.075
6.6 % (max acc.)	99.5 %	8.2 %	0.075
2.5 % (max J)	23.6 %	87.7 %	**0.113**

The canonical 6.5 % threshold is heavily skewed toward sensitivity at the expense of specificity. The balanced-optimal threshold on the consolidated benchmark is approximately 2.6 times tighter. Whether $J = 0.113$ constitutes a useful classifier is a separate question, and quantitatively it does not. The recalibration finding itself is, however, a direct and defensible output of treating the empirical rule as a diagnostic classifier rather than as a fixed cutoff. The full threshold sweep is shown in Figure 2, where the curve lies close to the no-skill diagonal.

4. Impact

The contribution consolidates an open benchmark that was not previously available in unified form, and supplies the diagnostic- classifier statistics that were not previously applied to the canonical high-entropy alloy rules in the open literature. Three classes of impact are anticipated.

The first is the calibration of expectations. The collapse of both binary rules to near-zero Youden's J on the consolidated benchmark serves as a quantitative caveat for downstream users of those rules. The 80 %-accuracy figures quoted in the original papers reflect smaller, different datasets and a different choice of evaluation metric. On a consolidated open dataset, the rules behave very differently.

The second is the provision of a substrate for subsequent methodological work. A candidate model, a CALPHAD-derived classifier, or a new empirical descriptor can now be scored against the same dataset and reported with the same statistical vocabulary as the four canonical rules. The result is a head-to-head comparison rather than incommensurable single-publication accuracies.

The third is accessibility. The Pyodide front end allows a researcher or student to enter a composition into a browser tab and to obtain the same numerical outputs as a local install of the Python package, with

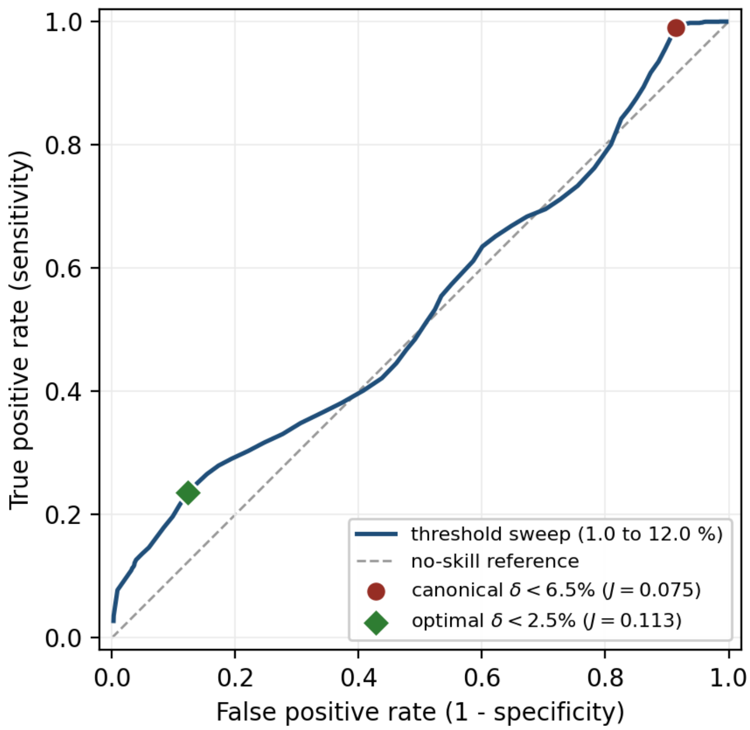

Figure 2: Receiver-operating-characteristic curve for the Zhang δ rule on the v0.1.0 benchmark.

the release wheel itself loaded into the browser rather than a separate JavaScript reimplementation. This removes the local-install barrier without compromising the library-as-canonical-implementation guarantee.

5. Conclusions

The article has presented HEA-BENCH 0.0.1, an open Python library and benchmark suite for high-entropy alloy phase prediction. The package ships a 7784-composition consolidated benchmark drawn from three primary open sources with explicit cross-source provenance and conflict tracking. It provides reference-baseline diagnostic-classifier evaluations of the four canonical empirical rules. It reports a receiver-operating-characteristic recalibration finding for the Zhang δ rule, with a dataset-optimal threshold of $\delta < 2.5\,\%$ rather than the canonical $\delta < 6.5\,\%$. It includes an in-browser Pyodide front end that runs the same release wheel as a local package-manager install. The package is released under the MIT License and is covered by more than 150 tests, including pinned regression tests on every reported benchmark statistic. The intent is for the artifact to serve as the substrate on which subsequent methods for high-entropy alloy phase prediction can be compared.

Planned future work has two strands. The first is an expansion of the elemental-property table to lift benchmark coverage from 86.7 % to approximately 95 %. The required additions are mechanical, because the corresponding pair-enthalpy entries are already present in the vendored Miedema table. The second is a v0.2.0 data release that exposes the cross-source label conflicts as a structured side-dataset for downstream disagreement-resolution work.

Conflict of interest

The author declares no competing financial or non-financial interests.

Acknowledgements

The author thanks the matminer maintainers for the open Miedema pair-enthalpy table, vendored under BSD-3-Clause, the Pyodide project for the browser-side Python runtime, and the authors of the three primary source datasets for releasing data under reusable terms. This manuscript was prepared with computer-assisted text generation. All numerical parameters, formulae, threshold values, and benchmark results were independently verified against the cited primary sources or computed from documented inputs by the author.

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